

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Shipped containerized **ECS/EKS** deployments via **GitHub Actions CI/CD** with per-node **Redis checkpointing** for fault-tolerant recovery, sustaining **99%+ uptime** across **millions of daily operations**.
- Architected cloud-native **distributed executor** routing pipeline nodes to **Docker** and **AWS Lambda**; designed an **S3** artifact store with **IaC**-backed deployment patterns.
- Built **distributed search** and retrieval infrastructure on **Elasticsearch** with **knowledge graph** enrichment, serving typed **REST APIs** at **sub-200ms p99** latency.
- Built **Prometheus/Grafana** observability and **autoscaling** for inference serving, plus **SSE**-based live monitoring with **automated incident detection**, cutting redundant API calls **90%**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** on **GCP Cloud Run** with **Kubernetes**-managed autoscaling; optimized distributed service throughput and cost efficiency.
- Built production **observability** with automated **CI/CD rollback** and **failure alerting**; on **on-call** rotation resolving distributed system incidents end-to-end.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and **SQL**-backed **distributed systems** for **1M+ users** at **99.9% uptime** on **AWS** (EC2, RDS, SQS); cut escalations **60%**.
- Built automated **deployment-gating** running quality, latency, and regression checks before production with **rollback** criteria; cut regressions **28%** and validation from **6 hours to 90 min**.
- Designed high-throughput **distributed messaging** pipelines on **AWS SQS**, cutting nightly batch aggregation from **8 minutes to 90 seconds** via indexed views and partitioning.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production **eval and observability platform** with **drift detection**, latency tracking, and alerting across versions; **FastAPI**, **React/TypeScript**, **PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Go, Java, C++, SQL, TypeScript, HTML/CSS

Infrastructure: Kubernetes, Docker, Terraform (IaC), AWS (EC2, S3, RDS, Lambda, EKS, SQS), GCP (Cloud Run)

Observability & Data: Prometheus, Grafana, Alerting, Elasticsearch, PostgreSQL, Redis, Kafka

Practices: Distributed Systems, CI/CD (GitHub Actions), On-call, RCA, Data Structures & Algorithms

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer